

A Computational Model of Merger as Phonetic Dispersion over Words

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LABPHON20

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Background: dispersion vs. merger

Dispersion Theory (Liljencrants & Lindblom, 1972): languages maximize perceptual contrast between speech sounds

- Succeeds in explaining common vowel systems
 - e.g. nearly all languages have corner Vs /i a u/ (Crothers, 1978)

Merger: sound change where two phonemes lose contrast

- e.g. *cot-caught* merger in North American English
- Requires lack of dispersion between phonemes

Problem: How to reconcile these seemingly opposite phenomena?

Insight: words matter

Ganong effect (1980): bias in segment perception toward real words

- But dispersion models have ignored word-level perception

Functional load hypothesis (Gillieron, 1918): merger more likely for contrasts distinguishing fewer words (i.e. minimal pairs)

- Supported statistically by sound change data (Wedel et al, 2013)

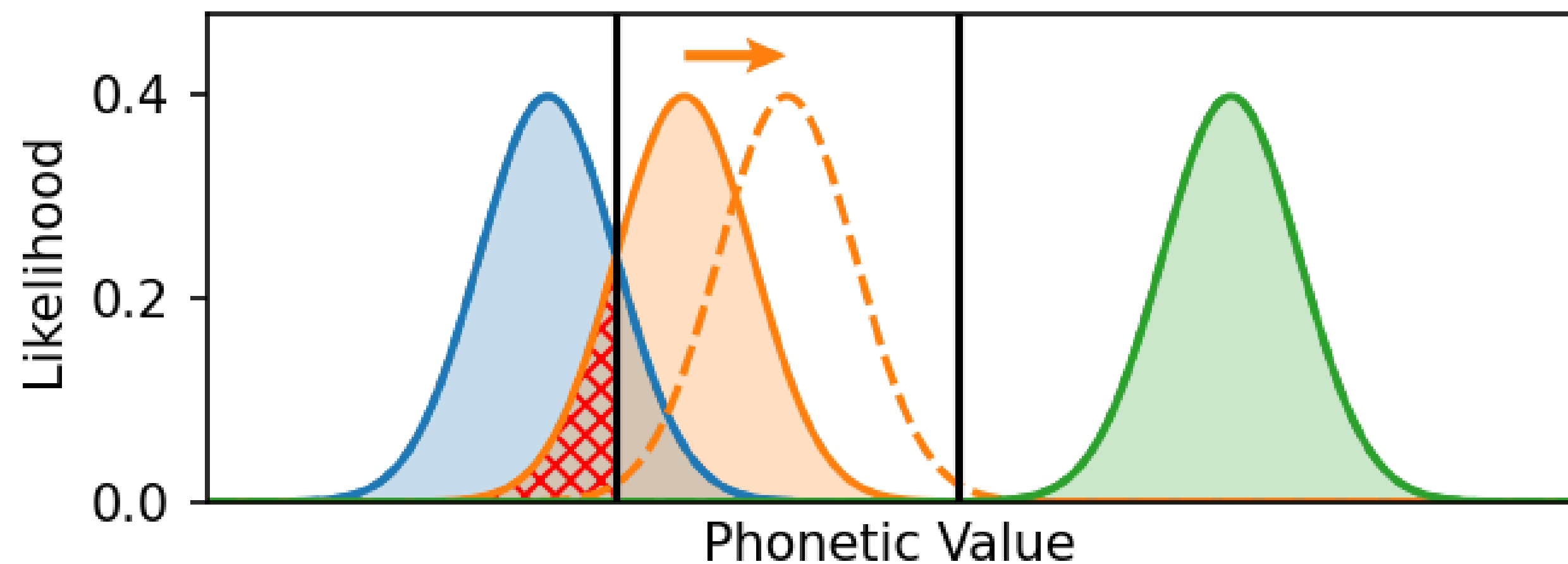
Idea: Maybe dispersion only matters for contrasting words?

Iterated Confusion Minimization (ICM)

Algorithm and model of emergent dispersion based on minimization of listener confusion (McGahay, 2025)

- Outperforms previous dispersion models (Liljencrants & Lindblom 1972; Schwartz et al., 1997) in predicting vowel system typology

Perceptual Optimization of Phonetic System



Maximum-likelihood listener: simplifies Bayesian models of optimal listeners (e.g. Clayards et al., 2008; Feldman et al., 2009)

- Sharp perceptual boundaries (vertical lines)
- Confusion on 'wrong' side of boundary (red cross-hatching)

Optimal speaker reduces confusion by adopting more dispersed distributions farther from boundaries (dashed distribution)

Then, a listener updates their perception and the process repeats

Proposal: ICM generalized over words

Approach: ICM simulation over toy lexicon of CVCs

- 9 /C_C/ contexts and 6 original vowels /i e ε a o u/
- Empirically informed V distributions over F1/F2 (McGahay, 2025)

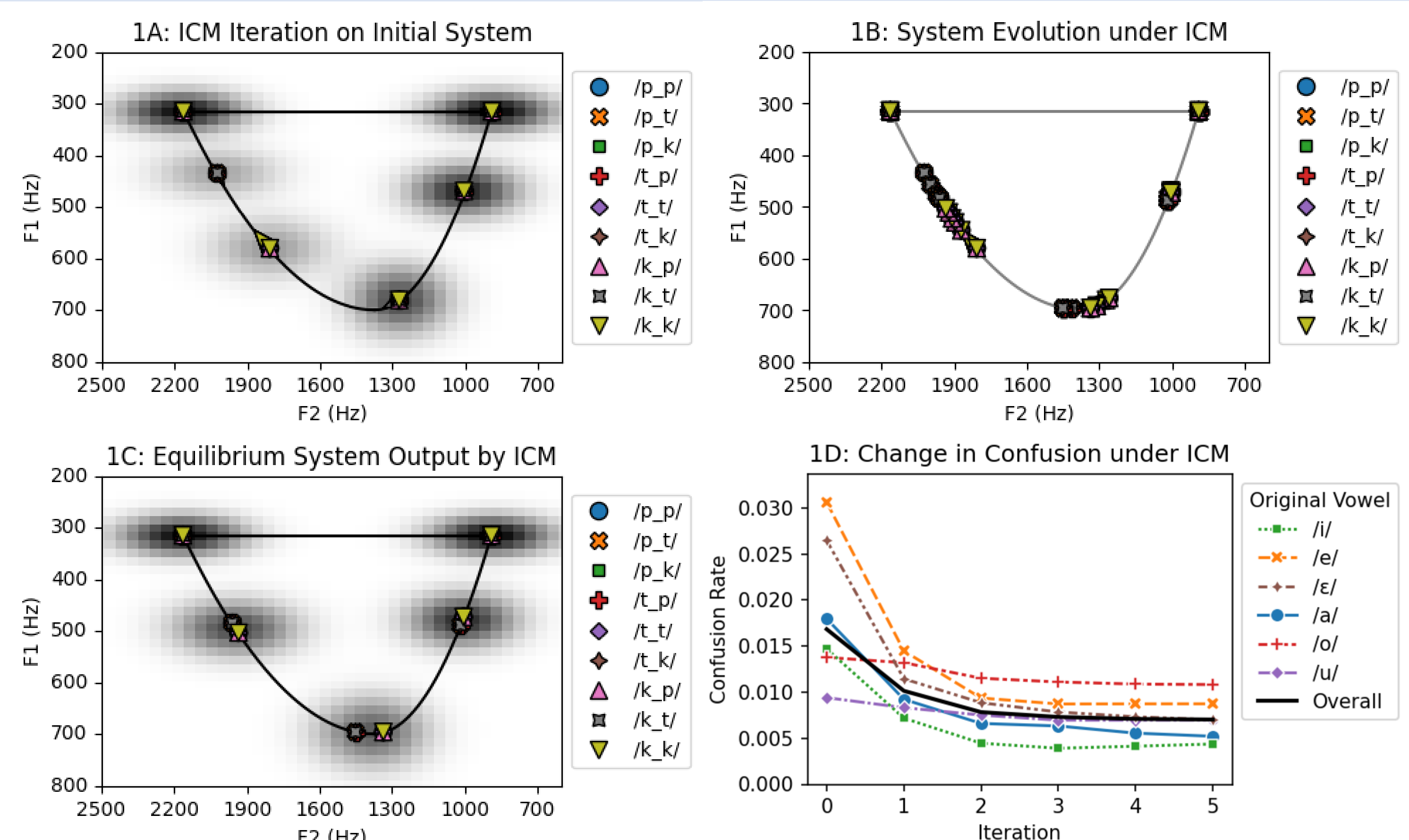
Simplifying assumption: /C_C/ contexts never confused

- Removes need to model consonant distributions
- Confusion limited to vowels within a minimal pair set

Simulations will allow vowels from same the phoneme class to take different phonetic distributions in different words

Simulation 1: contrast with no minimal pairs

Original /e/-/ε/ has no minimal pairs, other contrasts have 9

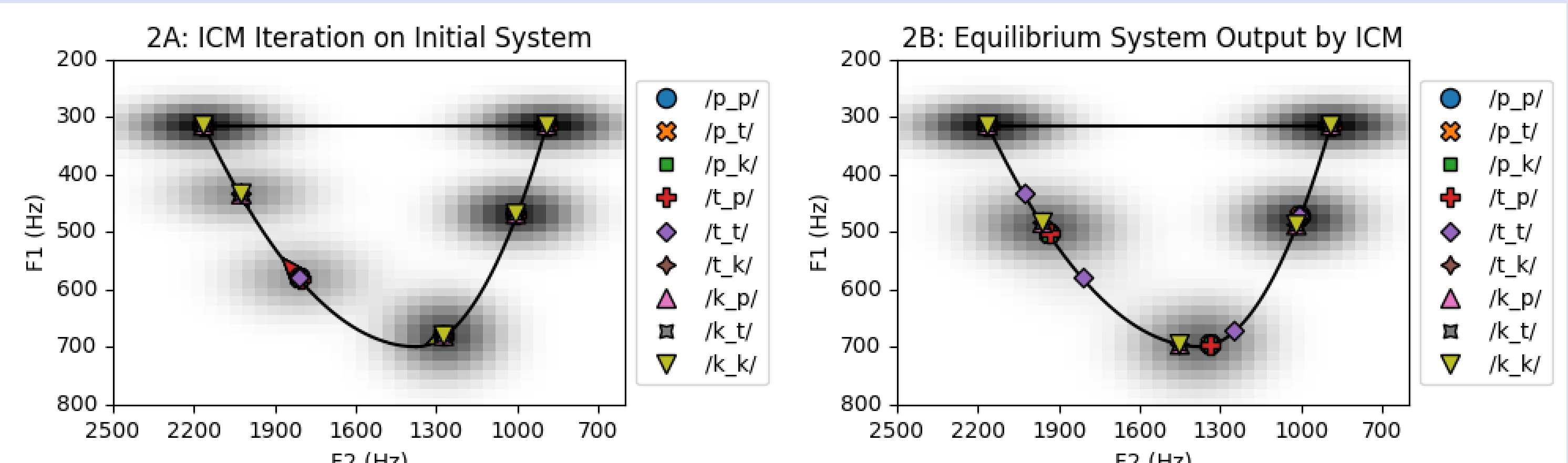


Initial system (1A) evolves (1B) until merger of /e/-/ε/ (1C), reducing overall confusion (black line of 1D)

- Phonetic space wasted on non-contrastive /e/-/ε/ reallocated to better distinguish /i/-/e/ and /ε/-/a/

Simulation 2: contrast with one minimal pair

Same as Simulation 1, but /e/-/ε/ now has a single minimal pair



/e/-/ε/ merge (2B) except in minimal pair /tet/-/tet/ (purple diamonds)

- merger avoidance to prevent homophony (cf. Blevins & Wedel, 2009)
- Difference from Sim. 1 consistent with finding that merger more likely with zero vs. at least one minimal pair (Wedel et al, 2013)

Despite 6 contrasting vowels, output 2B has only 5 phonetic clusters

- Difficulty for distributional learning? (e.g. Feldman et al., 2013)

Conclusion

Takeaway: applying dispersion principles to words predicts merger

- Segment contrasts that do not contrast (m)any words waste space since ambiguity is resolved by lexical knowledge (cf. Ganong, 1980)

Ongoing work:

- Combining ICM with a distributional learner (Feldman et al., 2013) to see if even full merger of minimal pairs can be predicted
- Using less idealized models to simulate real cases of merger
- Determining conditions under which merger vs. split predicted
 - ICM over sequences with confusable consonants predicts vowel allophony to cue difficult C contrasts (McGahay, 2026)

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